Crossing-Point Analysis for Weighted Rolling Validation: Theory

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1 Motivation and setup

Rolling validation (RV) sums validation losses over successive training sizes, while the desired model has lower population risk at the current size. Analysis 08 compared a sample mean with an oracle that was always preferable, so it had no model-preference crossing. Analysis 09 asks whether weighting later validation times can align the expected RV crossing with a population-risk crossing while retaining reliable selection near that crossing.

Let $Y_i = \theta + \xi_i$, where the ξ_i are independent $N(0,1)$ variables and $0 < |\theta| < 1$. Model 0 predicts with the training sample mean $\bar{Y}_n$; Model 1 always predicts zero. Both use squared loss. Write $R_k(n)$ for Model k's population prediction risk after n training observations and define

$$\Delta_R(n) = R_0(n) - R_1(n).$$

This is a current risk difference, evaluated only at size n. If $RV_{k,w}(n)$ is Model k's weighted sum of validation losses through the time with n training observations, define

$$\Delta_{\mathrm{RV},w}(n) = \mathbb{E}[RV_{0,w}(n) - RV_{1,w}(n)].$$

This is an accumulated expected difference. A negative difference favors Model 0. The risk and RV calculations are in Appendices A.1, A.2, and B.1.

2 Why ordinary RV crosses late

Theorem 2.1 (Population crossing)**.**

$$R_0(n) = 1 + \frac{1}{n}, \qquad R_1(n) = 1 + \theta^2, \qquad \Delta_R(n) = \frac{1}{n} - \theta^2, \qquad n_{\mathrm{pop}} = \lfloor \theta^{-2} \rfloor + 1.$$

Here n_{pop} is the first size at which $\Delta_R(n) < 0$.

Interpretation. Model 0 becomes preferable once its estimation variance falls below Model 1's squared bias. **Proof.** See Appendix A.1.

At validation time i, the training size is $i-1$; ordinary RV gives all validation times equal weight.

$$\Delta_{\mathrm{RV}}(n) = \sum_{i=2}^{n+1} \left(\frac{1}{i-1} - \theta^2 \right) = \sum_{i=2}^{n+1} \frac{1}{i-1} - n\theta^2, \qquad \sum_{i=2}^{n+1} \frac{1}{i-1} = \log n + O(1). \tag{1}$$

Thus the crossing is asymptotically determined by

$$\Delta_{\mathrm{RV}}(n) = \log n - n\theta^2 + O(1), \qquad \log n \approx n\theta^2.$$

Proposition 2.2 (Ordinary-RV lag)**.** Let n_{RV} be the first size with $\Delta_{\mathrm{RV}}(n) < 0$. As $|\theta| \to 0$,

$$n_{\mathrm{RV}} \sim \theta^{-2} \log(\theta^{-2}), \qquad n_{\mathrm{pop}} \sim \theta^{-2}, \qquad \frac{n_{\mathrm{RV}}}{n_{\mathrm{pop}}} \sim \log(\theta^{-2}) \to \infty.$$

Interpretation. Ordinary RV crosses asymptotically later because accumulated early validation contributions create an additional logarithmic delay. **Proof.** See Appendices A.2 and A.3.

3 Can weighting align the crossing?

Give validation time i a positive deterministic weight w_i independent of the data. The expected weighted difference is

$$\Delta_{\mathrm{RV},w}(n) = \sum_{i=2}^{n+1} w_i \left(\frac{1}{i-1} - \theta^2 \right) = \sum_{i=2}^{n+1} \frac{w_i}{i-1} - \theta^2 \sum_{i=2}^{n+1} w_i. \tag{2}$$

Since $\sum_{i=2}^{n+1} w_i > 0$, dividing by total weight does not change the crossing and puts weighted RV on an average scale comparable to population risk.

$$\frac{\Delta_{\mathrm{RV},w}(n)}{\sum_{i=2}^{n+1} w_i} = \frac{\sum_{i=2}^{n+1} w_i/(i-1)}{\sum_{i=2}^{n+1} w_i} - \theta^2 = r_n - \theta^2, \tag{3}$$

with

$$r_n = \frac{\sum_{i=2}^{n+1} w_i/(i-1)}{\sum_{i=2}^{n+1} w_i}.$$

The population-risk difference is

$$\Delta_R(n) = \frac{1}{n} - \theta^2.$$

Here r_n is the effective reciprocal training size seen by weighted RV, so weighted RV has the same form as population risk, with $1/n$ replaced by r_n. Since

$$r_n \geq \frac{1}{n},$$

expected weighted RV cannot cross before population risk. Alignment requires matching these reciprocal training sizes. Since

$$\frac{r_n}{1/n} = n r_n,$$

the following conditions are exactly equivalent:

$$n r_n \longrightarrow 1 \quad \Longleftrightarrow \quad \frac{r_n}{1/n} \longrightarrow 1 \quad \Longleftrightarrow \quad r_n \sim \frac{1}{n}. \tag{4}$$

The expectation and crossing order are proved in Appendices B.1 and B.2.

For large n, weighted RV therefore uses essentially the same reciprocal training size as population risk: their crossings are governed by $r_n \approx \theta^2$ and $1/n \approx \theta^2$, respectively. The limit $|\theta| \to 0$ puts both crossings in this large-n regime, since

$$n_{\mathrm{pop}} \sim \theta^{-2} \longrightarrow \infty \qquad (|\theta| \to 0).$$

Thus $r_n \sim 1/n$ applies at the moving crossing, giving the alignment stated below.

Theorem 3.1 (Sufficient crossing alignment)**.** If the positive deterministic weights do not depend on θ and $n_{\mathrm{RV},w}$ denotes the first size with $\Delta_{\mathrm{RV},w}(n) < 0$, then

$$nr_n \longrightarrow 1 \quad \Longrightarrow \quad n_{\mathrm{RV},w} \sim n_{\mathrm{pop}} \quad \Longleftrightarrow \quad \frac{n_{\mathrm{RV},w}}{n_{\mathrm{pop}}} \longrightarrow 1 \qquad (|\theta| \to 0),$$

with $n_{\mathrm{RV},w}$ existing for all sufficiently small nonzero θ.

Interpretation. Matching the latest reciprocal training size aligns the expected crossings. **Proof.** See Appendix B.4.

$$w_i = 1: \qquad r_n \sim \frac{\log n}{n}, \qquad nr_n \sim \log n \to \infty;$$

$$w_i = [\log(i+1)]^p: \qquad r_n \sim \frac{\log n}{(p+1)n}, \qquad nr_n \sim \frac{\log n}{p+1} \to \infty \qquad (p > 0).$$

$$w_i = \log\log(i+e): \qquad r_n \sim \frac{\log n}{n}, \qquad nr_n \to \infty;$$

$$w_i = q^i: \qquad r_n \sim \frac{1}{n}, \qquad nr_n \to 1 \qquad (q > 1).$$

Ordinary, log-power, and log-log weights retain too much historical influence for asymptotic alignment, while geometric weights concentrate sufficiently on recent validation times to align the expected crossing. The weight asymptotics are proved in Appendices B.5, B.6, and B.7.

4 Can alignment and concentration hold together?

The realized weighted-RV difference is

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i\{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}.$$

This is the random score whose expectation Section 3 studied, so

$$\mathbb{E}\Xi_{n,w} = \Delta_{\mathrm{RV},w}(n).$$

Section 3 established the normalized expectation; multiplying back by total weight recovers its mean:

$$\frac{\Delta_{\mathrm{RV},w}(n)}{\sum_{i=2}^{n+1} w_i} = r_n - \theta^2, \qquad \Delta_{\mathrm{RV},w}(n) = \mathbb{E}\Xi_{n,w} = \left(\sum_{i=2}^{n+1} w_i\right)(r_n - \theta^2).$$

An expected crossing alone does not ensure reliable realized selection: the score must stay close enough to its mean to preserve its sign. When $\mathbb{E}\Xi_{n,w} \neq 0$, a sufficient Chebyshev criterion is

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \to 0.$$

Substituting the recovered mean gives

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} = \frac{\mathrm{Var}(\Xi_{n,w})}{\left(\sum_{i=2}^{n+1} w_i\right)^2 (r_n - \theta^2)^2}.$$

Expanding the single-step loss difference produces the linear term $-2\theta\xi_i$; after weighting and summing over validation times, this becomes

$$-2\theta \sum_{i=2}^{n+1} w_i \xi_i.$$

Since the ξ_i are independent $N(0,1)$, its variance is

$$\mathrm{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i\right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2.$$

The full loss expansion and variance decomposition are given in Appendices D.1 and D.2.

The linear and quadratic noise are uncorrelated, so this contribution is a lower bound for the full variance:

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \geq \frac{4\theta^2 \sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2 (r_n - \theta^2)^2}.$$

Since the population crossing satisfies

$$n_{\mathrm{pop}} \sim \theta^{-2}, \qquad \frac{n}{n_{\mathrm{pop}}} \sim n\theta^2 \qquad (|\theta| \to 0),$$

$n\theta^2$ measures the current training size relative to that crossing. We therefore study

$$n\theta^2 \to c \in (0, \infty),$$

with $c < 1$ before the population crossing, $c > 1$ after it, and $c = 1$ at its asymptotic boundary (Appendix C.1). Under Section 3's alignment condition,

$$nr_n \to 1 \quad \Longleftrightarrow \quad r_n \sim \frac{1}{n}, \qquad \theta^2 \sim \frac{c}{n}, \qquad r_n - \theta^2 \sim \frac{1-c}{n} \quad (c \neq 1).$$

Substituting these scales into the lower bound gives

$$\frac{4\theta^2 \sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2 (r_n - \theta^2)^2} \sim \frac{4c}{(1-c)^2}\, n \frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2}.$$

Thus

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2}$$

determines the asymptotic size of the linear-noise contribution to the Chebyshev ratio near the population crossing, up to the fixed factor $4c/(1-c)^2$.

This quantity measures how concentrated the weights are, not where they are concentrated. Under equal weighting, all n validation times contribute equally and the quantity is 1. If the weight is concentrated on fewer validation times, less independent noise is averaged out and the quantity becomes larger.

The alignment condition $nr_n \to 1$ adds the location requirement: because r_n is a weighted average of

$$1, \frac{1}{2}, \ldots, \frac{1}{n},$$

making $r_n \sim 1/n$ forces the weight to move increasingly toward the latest validation times. Moreover, the validation times carrying most of the weight must occupy a vanishing fraction of all n validation times. Therefore the amount of effective averaging becomes smaller relative to n, forcing

$$n\frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2} \to \infty.$$

Hence the same late-time concentration that aligns the expected crossing also makes the linear noise increasingly large relative to the mean signal near that crossing.

Theorem 4.1 (Near-crossing obstruction)**.** Suppose the alignment condition

$$nr_n \to 1$$

holds. Then

$$n\frac{\sum_{i=2}^{n+1} w_i^2}{(\sum_{i=2}^{n+1} w_i)^2} \longrightarrow \infty.$$

Consequently, for any sequence satisfying

$$\theta \to 0, \qquad n\theta^2 \to c \in (0,\infty) \setminus \{1\},$$

the near-crossing Chebyshev ratio satisfies

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \longrightarrow \infty. \tag{5}$$

Interpretation. Aligning the expected crossing requires the weights to concentrate increasingly near the latest validation times. This weakens averaging, so near the moving crossing the random fluctuation becomes large relative to the mean signal rather than negligible. **Proof.** See Appendices C.2 and C.3; the near-crossing scaling and the exact boundary $c = 1$ are treated in Appendices C.1 and C.4.

5 Conclusion

What 09 establishes. Ordinary RV crosses too late because it accumulates earlier validation losses. Weighting can align the expected crossing, but under the 09 assumptions that alignment conflicts with near-crossing Chebyshev concentration.

What remains open. The result does not determine the exact limiting wrong-selection probability. Direct probability analysis or different validation designs, weight classes, or crossing models remain possible directions.

Appendix

A Population risk and ordinary RV

A.1 Population-risk formulas and strict crossing

We compute both population risks and locate their first strict crossing. The calculation separates estimation variance from squared bias.

Let $Y_{\text{new}} = \theta + \xi_{\text{new}}$ be independent of the training sample. Since $\bar{Y}_n = \theta + S_n/n$ with $S_n = \sum_{t=1}^{n} \xi_t$, the Model-0 error is $\xi_{\text{new}} - S_n/n$. Independence, zero means, and $\text{Var}(S_n) = n$ give $R_0(n) = 1 + 1/n$. Model 1 has error $\theta + \xi_{\text{new}}$, hence $R_1(n) = 1 + \theta^2$. Their difference is $1/n - \theta^2$, negative exactly when $n > \theta^{-2}$. The least integer strictly above θ^{-2} is $\lfloor \theta^{-2} \rfloor + 1$, including when θ^{-2} itself is an integer (the equality index is a tie). This proves Theorem 2.1.

A.2 One-time and ordinary-RV expectation

We derive the one-time expected loss difference and sum it to obtain the ordinary-RV mean. Independence of the fresh response makes the calculation direct.

At time $m+1$, $\bar{Y}_m = \theta + S_m/m$ and the fresh error of Model 0 is $\xi_{m+1} - S_m/m$; that of Model 1 is $\theta + \xi_{m+1}$. Expanding and using independence of ξ_{m+1} and S_m gives

$$\mathbb{E}[(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2] = (1 + 1/m) - (1 + \theta^2) = 1/m - \theta^2.$$

Summing over $m = 1, \ldots, n$ proves (1).

A.3 Ordinary-RV crossing asymptotics

We use harmonic-number bounds to locate the first strict negative expected RV difference.

Put $h = \theta^2$ and $N = h^{-1}$. The mean $H_n - nh$ has increments $1/(n+1) - h$, so it is initially increasing and eventually decreasing to $-\infty$; its first strict negative index exists and is unique. Integral bounds give $\log(n+1) \le H_n \le 1 + \log n$. To show that the crossing occurs near $N \log N$, we evaluate the expected difference at one point slightly below and one slightly above that scale. Fix $\varepsilon \in (0, 1)$ and set

$$n_- = \lfloor (1-\varepsilon) N \log N \rfloor, \qquad n_+ = \lceil (1+\varepsilon) N \log N \rceil.$$

Both tend to infinity; $\log n_\pm = \log N + \log\log N + O(1)$, while $n_\pm h = (1 \pm \varepsilon) \log N + o(1)$. Thus $H_{n_-} - n_- h = \varepsilon \log N + o(\log N) > 0$ and $H_{n_+} - n_+ h = -\varepsilon \log N + o(\log N) < 0$. The crossing lies between n_- and n_+. Divide by $N \log N$ and let $\varepsilon \downarrow 0$ to obtain $n_{\text{RV}} \sim N \log N$. Since $n_{\text{pop}} = N + O(1)$, Proposition 2.2 follows.

B Weighted RV and crossing alignment

B.1 Weighted expectation

For the appendix calculations, write $\Delta_w(n) = \Delta_{\text{RV},w}(n)$, $A_n = \sum_{i=2}^{n+1} w_i/(i-1)$, and $W_n = \sum_{i=2}^{n+1} w_i$, so $r_n = A_n/W_n$.

We derive the weighted expected RV difference from the one-time expectation. Linearity of expectation is enough, even though validation losses are dependent.

Apply the one-time expectation in Appendix A.2 with $m = i - 1$. Deterministic weights and linearity of expectation give

$$\Delta_w(n) = \sum_{i=2}^{n+1} w_i \big((i-1)^{-1} - \theta^2 \big) = \sum_{i=2}^{n+1} \frac{w_i}{i-1} - \theta^2 \sum_{i=2}^{n+1} w_i = A_n - \theta^2 W_n.$$

No independence between validation losses is asserted or needed.

B.2 Exact remainder and crossing order

We express the weighted RV comparison as current population risk plus a historical correction. Positivity of that correction gives the crossing order.

Divide the preceding identity by $W_n > 0$ to obtain (3); adding and subtracting $1/n$ also gives

$$\frac{\Delta_w(n)}{W_n} = \Delta_R(n) + \left(\frac{A_n}{W_n} - \frac{1}{n}\right).$$

Each $1/(i-1) \geq 1/n$. Their positive-weighted average therefore satisfies $A_n/W_n \geq 1/n$; for $n > 1$, at least one earlier term is strictly larger and has positive weight, so the inequality is strict. If $\Delta_R(n) \geq 0$, then $\Delta_w(n)/W_n \geq 0$. Consequently a strict weighted-RV crossing cannot precede n_{pop}. Equality is possible at $n = 1$ or if nonnegative (rather than positive) weights are supported only on the final term.

B.3 Monotonicity and crossing existence

We show that the weighted historical reciprocal average decreases as validation times are added. This identifies when its first strict crossing exists.

Write $r_n = A_n/W_n$. Adding the next validation time gives

$$r_{n+1} = \frac{W_n r_n + w_{n+2}/(n+1)}{W_n + w_{n+2}}.$$

Every term in r_n is at least $1/n > 1/(n+1)$, so $r_{n+1} < r_n$ for positive weights. Since r_n is strictly decreasing, once $r_n < h$ occurs it remains below h. Therefore the first strict crossing, when it exists, is unique. It exists exactly when $\inf_n r_n < h$; since r_n decreases, this is equivalent to $\lim_n r_n < h$. If the limit equals h, strict crossing need not occur. Under (4), $r_n \sim 1/n \to 0$, so crossing exists for every fixed $h > 0$; $r_1 = 1 > h$ because $0 < h < 1$.

B.4 Sufficient-alignment theorem

The proof brackets the weighted-RV crossing between training sizes slightly below and slightly above the population crossing.

Let $h = \theta^2 \downarrow 0$ and fix $\varepsilon \in (0, 1)$. Set $n_- = \lfloor (1-\varepsilon)/h \rfloor$ and $n_+ = \lceil (1+\varepsilon)/h \rceil$. Both grow, and (4) gives $r_{n_\pm} = (1 + o(1))/n_\pm$. Therefore

$$\frac{r_{n_-}}{h} \to \frac{1}{1-\varepsilon} > 1, \qquad \frac{r_{n_+}}{h} \to \frac{1}{1+\varepsilon} < 1.$$

By Appendix B.3, the first strict crossing lies in $(n_-, n_+]$ for small h. Rounding changes either endpoint by at most one. Divide by h^{-1} and let $\varepsilon \downarrow 0$; since $n_{\text{pop}} = h^{-1} + O(1)$, the crossing ratio tends to one. This proves Theorem 3.1.

B.5 Log-power weights and their crossings

We first derive A_n and W_n for log-power weights, then use their ratio to locate the expected crossing.

Let $w_i = [\log(i+1)]^p$ with fixed $p \geq 0$. A shift by one changes $\log(i+1)$ by $O(1/i)$, so its pth power has the same leading order as $(\log i)^p$ (with $p = 0$ interpreted as 1). Integral comparison, or summation by parts, yields

$$A_n \sim \int_2^n \frac{(\log x)^p}{x}\, dx = \frac{(\log n)^{p+1}}{p+1}.$$

For completeness, the summand is eventually decreasing: its logarithmic derivative is $(p/\log x - 1)/x < 0$ for $x > e^p$. Thus its sum differs from the integral by a bounded initial contribution and an error no larger than a fixed eventual first term. The shift error is summable for $p \geq 0$. Also, for any $\delta \in (0,1)$, the last $(1-\delta)n$ indices have weights $(\log n)^p(1+o(1))$ uniformly, while the first δn contribute at most $\delta n(\log n + O(1))^p$. Taking $n \to \infty$ and then $\delta \downarrow 0$ gives $W_n \sim n(\log n)^p$. Hence

$$r_n \sim \frac{\log n}{(p+1)n}.$$

For the crossing let $N = h^{-1}$ and $g_N = N \log N/(p+1)$. At $n_\pm = \lfloor (1 \mp \varepsilon) g_N \rfloor$ and $\lceil (1 \pm \varepsilon) g_N \rceil$, respectively, $\log n_\pm / \log N \to 1$, so $r_{n_\pm}/h \to (1 \mp \varepsilon)^{-1}$. Monotonicity from Appendix B.3 brackets the first crossing between them. Letting $\varepsilon \downarrow 0$ gives $n_{\mathrm{RV},w} \sim g_N$ and, since $n_{\mathrm{pop}} \sim N$, the ratio stated in the main text. This includes $p = 0$.

B.6 Log-log weight

We estimate A_n and W_n for the slower-growing log-log weight, then determine its crossing scale.

Set $f(x) = \log\log(x+e) > 0$. Its derivative is $f'(x) = 1/[(x+e)\log(x+e)]$. For large x, $f(x) \sim \log\log x$ and $f(x)/x$ is decreasing, so integral comparison applies. The shift from $f(x)$ to $\log\log x$ produces an integrable error after division by x: the mean-value theorem bounds it by $C/(x^2 \log x)$ for sufficiently large x. Substituting $u = \log x$,

$$A_n = \int^n \frac{\log\log x}{x}\, dx + O(1) = \log n \log\log n - \log n + O(1) \sim \log n \log\log n.$$

The same last-fraction argument as in Appendix B.5, using $f(\delta n)/f(n) \to 1$ for fixed $\delta > 0$, gives $W_n \sim n \log\log n$. Hence $r_n \sim (\log n)/n$. Bracketing at $n_\pm = (1 \pm \varepsilon) N \log N$ exactly as in Appendix B.5 gives $n_{\mathrm{RV},w} \sim N \log N$ and $n_{\mathrm{RV},w}/n_{\mathrm{pop}} \sim \log N$.

B.7 Geometric-weight alignment and effective averaging

We show that geometric weights align the expected crossing and calculate how concentrated their normalized weights remain.

For $w_i = q^i$ with $q > 1$, write $i = n + 1 - j$. Division by W_n makes the lag j have probabilities proportional to q^{-j}, $j = 0, \ldots, n-1$. The denominator $\sum_{j=0}^{n-1} q^{-j} \to q/(q-1)$ and the numerator for nr_n is

$$\sum_{j=0}^{n-1} q^{-j} \frac{n}{n-j}.$$

For $j \leq n/2$, $n/(n-j) \leq 2$ and tends to 1 for each fixed j; dominated convergence for the summable geometric series gives the same limit $q/(q-1)$. For $j > n/2$, the sum is at most $n \sum_{j>n/2} q^{-j} \to 0$. Thus $nr_n \to 1$. Direct finite geometric sums also give

$$\frac{\sum_{i=2}^{n+1} q^{2i}}{(\sum_{i=2}^{n+1} q^i)^2} \longrightarrow \frac{q-1}{q+1}.$$

Theorem 3.1 supplies the crossing ratio; the displayed square-weight limit measures its bounded effective averaging.

C Near-crossing alignment–concentration obstruction

C.1 The near-crossing scaling

We relate $n\theta^2$ to the moving population crossing and establish the before/after sign interpretation.

Put $h = \theta^2$. Theorem 2.1 gives $n_{\text{pop}} = h^{-1} + O(1)$. Thus $nh \to c$ places n at asymptotic fraction c of the population crossing. Moreover $n\Delta_R(n) = 1 - nh \to 1 - c$, proving the before/after sign interpretation for $c \neq 1$.

C.2 Weight concentration and effective sample size

The alignment condition forces asymptotically almost all normalized weight toward the latest validation times. We first show this concentration, then use Cauchy–Schwarz to show that the effective amount of averaging must be small.

Let $p_{n,i} = w_i/W_n$, so $\sum_i p_{n,i} = 1$. Alignment is

$$\sum_{i=2}^{n+1} p_{n,i} \left(\frac{n}{i-1} - 1 \right) \to 0.$$

All summands are nonnegative. For fixed $\varepsilon \in (0,1)$ and $i - 1 \leq (1-\varepsilon)n$, the factor is at least $\varepsilon/(1-\varepsilon)$. Hence the total mass there is at most $(1-\varepsilon)/\varepsilon$ times the displayed sum, and tends to zero. The remaining set has at most $\varepsilon n + 1$ indices and mass tending to one. Cauchy–Schwarz on that set gives

$$\sum_i p_{n,i}^2 \geq \frac{\left(\sum_{i-1>(1-\varepsilon)n} p_{n,i}\right)^2}{\varepsilon n + 1}, \quad \text{so} \quad \liminf_n n \sum_i p_{n,i}^2 \geq \frac{1}{\varepsilon}.$$

For any $K > 0$, choose $\varepsilon < 1/K$; the lower limit is then at least K. Thus $n \sum_i p_{n,i}^2 \to \infty$.

C.3 Proof of the obstruction theorem

We now combine the forced weight concentration from Appendix C.2 with the new linear-noise variance lower bound.

Under alignment, $r_n = (1 + o(1))/n$. If $nh \to c \neq 1$, then by (2)

$$\frac{\mathbb{E}\Xi_{n,w}}{W_n} = r_n - h = \frac{1 - c + o(1)}{n}.$$

Appendix D.2 proves that the quadratic variance is nonnegative and the additional linear variance equals $4h \sum_i w_i^2$. Therefore

$$\frac{\operatorname{Var}(\Xi_{n,w})}{(\mathbb{E}\Xi_{n,w})^2} \geq \frac{4h \sum_i p_{n,i}^2}{(r_n - h)^2} = \frac{4(c + o(1))\, n \sum_i p_{n,i}^2}{(1 - c + o(1))^2} \to \infty$$

by Appendix C.2. The mean is positive when $c < 1$ and negative when $c > 1$, for sufficiently large n. The proof establishes failure of a sufficient Chebyshev argument, not a positive lower bound on wrong-selection probability. This proves Theorem 4.1.

C.4 The exact boundary

The case $c = 1$ needs separate treatment because the leading mean vanishes there. We check what happens when the smaller-order mean is zero or nonzero.

If $nh \to 1$ under alignment, $r_n - h = o(1/n)$. When this difference is zero, the variance-to-mean-squared ratio is undefined. When it is nonzero, the preceding lower bound has numerator $4h\sum p_{n,i}^2$ and denominator $o(n^{-2})$; writing $r_n - h = \delta_n/n$ with $\delta_n \to 0$ gives a lower bound $4(1 + o(1))n\sum p_{n,i}^2/\delta_n^2 \to \infty$. No leading-order population sign is assigned at $c = 1$.

D Exact variance and a fixed-parameter comparison

D.1 Loss expansion and centering

We expand one validation loss difference and identify the new linear term caused by $\theta \neq 0$. Centering then isolates the quadratic and linear random parts.

Write $S_m = \sum_{t=1}^m \xi_t$. Since $\bar{Y}_m = \theta + S_m/m$,

$$(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2 = (\xi_{m+1} - S_m/m)^2 - (\theta + \xi_{m+1})^2 = \frac{S_m^2}{m^2} - 2\xi_{m+1}\frac{S_m}{m} - 2\theta\xi_{m+1} - \theta^2.$$

The first quadratic term has mean $1/m$ and the other random terms have mean zero. Therefore, with

$$Q_{n,w} = \sum_{m=1}^n w_{m+1}\left\{\frac{S_m^2 - m}{m^2} - 2\xi_{m+1}\frac{S_m}{m}\right\},$$

we have $\Xi_{n,w} - \mathbb{E}\Xi_{n,w} = Q_{n,w} - 2\theta\sum_{i=2}^{n+1} w_i\xi_i$. This also independently verifies (2).

D.2 Gaussian orthogonality and coefficients

We collect the quadratic coefficients and use Gaussian orthogonality to obtain the exact variance. This separates quadratic noise from the new linear-noise variance.

Expand $S_m^2 - m = \sum_{t=1}^m(\xi_t^2 - 1) + 2\sum_{1\le t<u\le m}\xi_t\xi_u$. Collecting each centered square gives coefficient

$$a_t = \sum_{m=t}^n \frac{w_{m+1}}{m^2}, \qquad 1 \le t \le n.$$

For each pair $t < l$, the square expansion contributes $2\sum_{m=l}^n w_{m+1}/m^2$; the validation term at $m = l - 1$ contributes $-2w_l/(l-1)$. Thus, with empty sums zero,

$$b_l = \sum_{m=l}^n \frac{w_{m+1}}{m^2} - \frac{w_l}{l-1}, \qquad 2 \le l \le n+1,$$

and

$$Q_{n,w} = \sum_{t=1}^n a_t(\xi_t^2 - 1) + 2\sum_{l=2}^{n+1} b_l \sum_{t=1}^{l-1}\xi_t\xi_l.$$

Independent standard Gaussians satisfy $\mathbb{E}(\xi_t^2 - 1)^2 = 2$ and $\mathbb{E}(\xi_t\xi_l)^2 = 1$ for $t \neq l$. Distinct centered squares, distinct unordered pair products, and a centered square against a pair product have covariance zero: in every cross-product, some independent centered Gaussian appears to an odd power or as a mean-zero factor. Each quadratic monomial also has zero covariance with every

ξ_i, because every centered third Gaussian moment is zero. Hence the quadratic and linear parts are orthogonal, and

$$\mathrm{Var}(Q_{n,w}) = 2\sum_{t=1}^{n} a_t^2 + 4\sum_{l=2}^{n+1}(l-1)b_l^2, \qquad \mathrm{Var}\left(-2\theta\sum_{i=2}^{n+1} w_i\xi_i\right) = 4\theta^2\sum_{i=2}^{n+1} w_i^2.$$

Adding proves the exact finite-n variance identity

$$\mathrm{Var}(\Xi_{n,w}) = 2\sum_{t=1}^{n} a_t^2 + 4\sum_{l=2}^{n+1}(l-1)b_l^2 + 4\theta^2\sum_{i=2}^{n+1} w_i^2. \tag{6}$$

D.3 Slow-weight variance orders

Step 1: Goal and weight class. We determine the quadratic variance order for the unweighted, log-power, and log-log families, then combine it with the linear variance to obtain their Chebyshev-ratio order.

Here $h = \theta^2$ and $nh \to c \in (0,\infty)$; consider $w_{m+1} = f(m)$, where $f(m) = 1$, $[\log(m+2)]^p$ for fixed $p > 0$, or $\log\log(m+1+e)$. In each case $f(n)$ is slowly varying, and splitting sums at δn as in Appendix B shows

$$\sum_{m=1}^{n} f(m)^2 \sim nf(n)^2.$$

Step 2: Control the a_t-part. We verify the quadratic order without borrowing it from 08. The coefficients a_t are bounded by $\sum_{m=t}^{\infty} f(m)/m^2$. For the listed f, integral comparison and eventual monotonicity of $f(x)/x^{3/4}$ give $\sum_{m=t}^{\infty} f(m)/m^2 \le Cf(t)/t$ for large t. Consequently $\sum_t a_t^2 \le C + C\sum_t f(t)^2/t^2 = O(1)$. Thus the a_t-part stays bounded.

Step 3: Rewrite the b_l-part. Set $T_n = \sum_{m=n+1}^{\infty} f(m)/m^2$ and $B_l = \sum_{m=l}^{\infty} f(m)/m^2 - f(l-1)/(l-1)$, so $b_l = B_l - T_n$ for $2 \le l \le n$ and $b_{n+1} = -f(n)/n$ (the harmless one-step indexing changes the following estimates only by lower-order terms). This separates the terminal contribution T_n from the residual B_l.

Step 4: Bound B_l. Integral comparison, summation by parts, and the mean-value theorem give

$$T_n \sim f(n)/n, \qquad |B_l| \le C\left(\frac{\sup_{x\in[l-1,l+1]}|f'(x)|}{1} + \frac{f(l)}{l^2}\right)$$

for smooth extensions of the listed f. To see the derivative bound, subtract the constant-$f(l)$ sum, whose discrepancy from $f(l)/(l-1)$ is $O(f(l)/l^2)$; summation by parts bounds the remaining variation by $C\int_l^{\infty}|f'(x)|x^{-1}dx$. This is $O((\log l)^{p-1}/l)$ for log powers and $O(1/(l\log l))$ for log-log; the constant case has zero derivative. Therefore $D_n = \sum_{l=2}^{n}(l-1)B_l^2$ is $O(1)$ for $f = 1$ and log-log, while for log powers it is bounded by

$$C + C\sum_{l=3}^{n}\frac{(\log l)^{2p-2}}{l} = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log\log n), & p = 1/2, \\ O((\log n)^{2p-1}), & p > 1/2. \end{cases}$$

The three cases follow by $u = \log x$ in the comparison integral $\int^n (\log x)^{2p-2}dx/x$. In every case $D_n = o(f(n)^2)$ when $f(n) \to \infty$; for $f = 1$, $D_n = O(1)$ and the exact $B_l = O(l^{-2})$ is handled below.

Step 5: Identify the dominant terminal contribution. The terminal contribution is

$$4T_n^2 \sum_{l=2}^{n} (l-1) \sim 4\frac{f(n)^2}{n^2}\frac{n^2}{2} = 2f(n)^2.$$

Step 6: Control the cross term. Expanding $(B_l - T_n)^2$, Cauchy–Schwarz bounds the magnitude of its cross term by $2\{D_n T_n^2 \sum_{l=2}^{n}(l-1)\}^{1/2} = o(f(n)^2)$ for the growing weights. The end coefficient contributes $4n(f(n)/n)^2 = o(f(n)^2)$. Thus $\mathrm{Var}(Q_{n,w}) \sim 2f(n)^2$ for the growing log and log-log weights. For $f = 1$, direct integral comparison yields $B_l = O(l^{-2})$. Thus $D_n = O(1)$ and $\sum_{l=2}^{n}(l-1)|B_l| = O(\log n)$. Since $T_n = O(1/n)$, the cross term is $O(\log n/n) = o(1)$; the end coefficient is $O(1/n)$. Therefore $\mathrm{Var}(Q_{n,w}) \sim 2$ also in the unweighted case.

Step 7: Add the linear variance. The linear variance is $4h\sum_i w_i^2 \sim 4cf(n)^2$. Hence the full variance is of order $f(n)^2$ (indeed asymptotic to $(2+4c)f(n)^2$ for the growing weights).

Step 8: Conclude the Chebyshev-ratio order. Meanwhile Appendix B gives $\mathbb{E}\Xi_{n,w} \sim (\log n)f(n)/(p+1)$ for log powers and $\sim (\log n)f(n)$ for log-log; the $-hW_n$ term is only $O(f(n))$. The ordinary-weight mean is $\sim \log n$ and its full variance tends to $2 + 4c$. Therefore all four Chebyshev ratios have order $(\log n)^{-2}$. Their means are positive for fixed c, establishing the sign statement in the main text.

D.4 Stretched-exponential weights at fixed θ

We verify three properties in order: crossing alignment, vanishing normalized weight concentration, and fixed-θ concentration.

Let $w_i = \exp(i^\beta)$ with $0 < \beta < 1$. Put $F(x) = e^{x^\beta}$. Substitution $u = x^\beta$ and integration by parts yield, for $a = 1, 2$,

$$\int^n F(x)^a\,dx = \frac{1}{\beta}\int^{n^\beta} e^{au}u^{1/\beta-1}\,du \sim \frac{n^{1-\beta}e^{an^\beta}}{a\beta}.$$

Because F is increasing, bounding sums between adjacent integrals changes them by at most $O(F(n)^a)$, negligible relative to $n^{1-\beta}F(n)^a$. Endpoint shifts by one are negligible since $(n+1)^\beta - n^\beta \to 0$. Therefore

$$W_n \sim \frac{n^{1-\beta}e^{n^\beta}}{\beta}.$$

For any fixed $\delta > 0$, the total weight at $i - 1 \le (1-\delta)n$ is at most $ne^{((1-\delta)n+O(1))^\beta} = o(W_n)$, because the exponential gap is of order n^β. On the remaining indices, $n/(i-1) \le 1/(1-\delta)$; for the earlier indices their weighted contribution to nr_n is negligible by the same exponential gap even after the factor $n/(i-1) \le n$. Hence $1 \le \liminf nr_n \le \limsup nr_n \le 1/(1-\delta)$, and $\delta \downarrow 0$ proves alignment.

The same sum comparison gives

$$\sum_i w_i^2 \sim \frac{n^{1-\beta}e^{2n^\beta}}{2\beta}, \qquad \frac{\sum_i w_i^2}{W_n^2} \sim \frac{\beta}{2n^{1-\beta}} \to 0.$$

These sums show an effective terminal window of order $n^{1-\beta} = o(n)$.

For fixed $\theta \neq 0$, $\mathbb{E}\Xi_{n,w}/W_n = r_n - \theta^2 \to -\theta^2$. To bound the quadratic variance, define the centered single-time quadratic increment $U_m = (S_m^2 - m)/m^2 - 2\xi_{m+1}S_m/m$. Gaussian moments and independence give $\|U_m\|_2 \leq \sqrt{2}/m + 2/\sqrt{m} \leq C/\sqrt{m}$. Minkowski's inequality and then Cauchy–Schwarz yield

$$\left\| \frac{Q_{n,w}}{W_n} \right\|_2 \leq C \sum_{m=1}^{n} \frac{w_{m+1}}{W_n \sqrt{m}} \leq C \left(\sum_{m=1}^{n} \frac{w_{m+1}}{W_n m} \right)^{1/2} = C\sqrt{r_n} \to 0.$$

The normalized linear variance is $4\theta^2 \sum_i w_i^2/W_n^2 \to 0$. Orthogonality from Appendix D.2 then gives $\mathrm{Var}(\Xi_{n,w})/W_n^2 \to 0$, and division by the mean-squared limit θ^4 proves fixed-θ concentration. This does not apply to $\theta \to 0$ with $n\theta^2 \to c$.